

12 Uniform Continuity

Let $f : A \rightarrow \mathbb{R}$ be a function on a set $A \subseteq \mathbb{R}$. Recall that continuity at a point c allows the choice of δ to depend on both ε and c . Uniform continuity strengthens this by requiring a single δ that works across the entire domain.

Definition 0.1. A function $f : A \rightarrow \mathbb{R}$ is uniformly continuous on A if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for all $x, y \in A$,

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Compare this with the pointwise definition, where δ may vary with c . The key theorem below shows compactness is exactly what's needed to upgrade continuity to uniform continuity.

Theorem 0.1 (Heine–Cantor). *If $f : K \rightarrow \mathbb{R}$ is continuous and K is compact, then f is uniformly continuous on K .*

A quick numerical check — verifying the δ/ε bound on a sample function:

```
def is_uniformly_cont(f, eps, xs):  
    for delta in grid:  
        if all(abs(f(x) - f(y)) < eps  
               for x, y in pairs(xs, delta)):  
            return delta
```

On the grid tested here, the search returns $\delta = 0.02$, well within the tolerance we assumed at the start of the lecture. Note that this value depends on the grid resolution, not on any particular point c — consistent with the definition above.